

Project Lasers: Comprehensive Technical Review

Verification Status and Detailed Strategic Action Plan

1. Executive Summary

This document provides a comprehensive technical review of Project Lasers, a self-consistent multiphysics EDA simulator for telecommunication diode lasers (1.31 μm / 1.55 μm). We evaluate the successfully verified 2D-1D multiscale wave-optical coupling loop, the implementation of localized active gain and lattice heat transport models, and specify the detailed technical roadmap for the next development phases. The model manual v2.3 (semiconductor_model_v2.pdf) serves as the exact physical specification for these solvers.

2. Verified Solver Architectures

The simulator successfully integrates and converges the following solver modules:

1. CarriersSolver: Solves Poisson and drift-diffusion current densities (J_n and J_p) using the FVM formulation. The Jacobian matrix is scaled by $1.0\text{e-}26$, and carrier concentration bounds are restricted to prevent numerical overflow, ensuring robust convergence in the dogleg nonlinear subsolver (CoDoSol).
2. EMSolver (Unified Wave Optics): Solves the complex 2D transverse Helmholtz equation. Spontaneous emission is coupled as a localized current density source (J_{sp}), and stimulated emission is driven by the spatial overlap between the mode intensity $|E_{2d}(y,z)|^2$ and the active region gain $g(y,z,\omega)$. The separate PhotonSolver has been retired, ensuring that local carrier depletion (spatial hole burning) is physically consistent with the optical mode confinement.
3. ThermoSolver: Governs lattice heat conduction driven by Joule, recombination, and optical heating sources. Thermal convergence is stabilized by applying a Steady State Relaxation Factor of 0.5, preventing numerical oscillations under high forward bias.
4. 1D Longitudinal Solver: Calibrates the 2D transverse cavity escape loss ($\alpha_{\text{escape}} = 24.04 \text{ cm}^{-1}$) by analytically solving the traveling wave intensity profiles along the cavity axis (x) and feeding the midpoint ratio back to tune the SIF Photon Decay Rate ($1.92333\text{e}11 \text{ s}^{-1}$).

3. Detailed Strategic Action Plan (What to Do Next)

To advance the simulator from its current steady-state coupled baseline toward a commercial-grade, high-fidelity Electronic Design Automation (EDA) tool, we define four critical development tasks:

Task 1: Transition to Curl-Conforming Edge Elements in EMSolver

- Objective: Eliminate non-physical spurious modes and represent interface boundary conditions correctly.
- Physics/Math: Ridge-waveguide lasers contain sharp index steps at material boundaries (e.g., active core to InP cladding). Standard nodal vector solvers (E_x , E_y components solved at nodes) struggle with these steps, violating normal field discontinuity and tangential continuity.
- Implementation: Re-formulate the EMSolver bulk assembly to use Elmer's native $H(\text{curl})$ edge-element basis functions. This mathematically guarantees the continuity of tangential electric fields across material boundaries and suppresses spurious solutions, providing clean, single-mode eigenvalues.

Task 2: Automated Physics-Informed Neural Network (PINN) Calibration

- Objective: Dynamically calibrate active region recombination and gain parameters using experimental data.
- Physics/Math: Parameters like the spontaneous radiative coefficient (B_{sp}), Auger coefficients (C_n , C_p), and transition energy shift coefficients vary depending on material growth and doping quality.
- Implementation: Connect Elmer's diagnostic outputs (field ranges, terminal voltage, and output power) to the neural network in the 'pinn/' folder. Build a closed-loop optimization script (using PyTorch or SciPy) that compares simulated P-I-V curves against measurements and backpropagates errors to tune the SIF coefficients automatically.

Task 3: Dynamic Poynting Flux Mode Normalization

- Objective: Replace the static scaling factor (1.0e10) with a self-consistent physical conversion factor.
- Physics/Math: The scaling between electromagnetic wave intensity $|E|^2$ and photon density s_{ph} must depend on the physical energy transport velocity (group velocity v_g) and Poynting power flux.
- Implementation: Write an internal Fortran function in EMSolver to compute the overlap integral: $P_{norm} = \iint |E_{2d}(y,z)|^2 dy dz$. Compute the local group velocity v_g and use the Poynting vector flux to scale the output mode profile dynamically, matching the actual lased cavity power P_{cav} .

Task 4: Multi-Mode and Spectral Lasing Resolution

- Objective: Simulate spectral broadening, side-modes, and side-mode suppression ratio (SMSR).
- Physics/Math: High-power diode lasers operate in multiple longitudinal modes. Dynamic gain-profile changes cause mode competition, shifting the emission spectrum.
- Implementation: Extend EMSolver to solve for the first N eigenvalues (multiple transverse modes) and map them to discrete Fabry-Perot cavity resonance modes. Incorporate a spectral gain loop inside the longitudinal solver to track the power distribution of individual side-modes, allowing spectral line-shape simulations.

Status: Technical Review & Roadmap Finalized

Release Version: 2.1 (Decoupled Multiscale Wave-Optical)

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